Assignment 8 - Task List, Milestones, Timeline, and Effort Matrix

BarCat Deals - CS 5001 Senior Design I / CS 5002 implementation planning.

Advisor: Dr. Hrishikesh Vinayak Bhide.

Team members: Brian Nguyen, Kaustubh Mathur, Ved Sanap, Meredith Bartel, Kiki Vasilev.

Task List

No.	Task	Primary owner
01	Specify the C1 Student Web App and G2 Notifications requirements and technical specifications with acceptance traceability.	Brian Nguyen
02	Specify the C2 Owner Web Portal price policy and test/validation plan for seven-item submissions and publication timing.	Kaustubh Mathur
03	Refine the C3 Backend API Service and G1 Authentication and authorization contracts, edge-case coverage, and ethics analysis.	Ved Sanap
04	Document the C4 Deal Database and G4 Logging and health recovery plans, technical references, and technical communication package.	Meredith Bartel
05	Investigate C5 Deal Ingestion Service, C6 External Social Media APIs, and G3 Configuration access and release gates for the fall presentations.	Kiki Vasilev
06	Develop the C4 Deal Database migrations, relational invariants, and representative venue and deal fixtures.	Meredith Bartel
07	Develop the C3 Backend API Service shared data contracts, validation errors, and I1-I4 contract test harness.	Ved Sanap
08	Develop the C1 Student Web App Flask/Jinja feed and detail templates using fixture-backed data.	Brian Nguyen
09	Develop the C2 Owner Web Portal login screens and seven-item submission form with accessible field validation.	Kaustubh Mathur
10	Develop the C5 Deal Ingestion Service normalized C6 External Social Media APIs adapters and replayable post fixtures.	Kiki Vasilev
11	Develop C3 Backend API Service and G1 Authentication and authorization guest age sessions and verified owner access.	Ved Sanap

No.	Task	Primary owner
12	Develop C1 Student Web App live feed integration, pagination, source and freshness displays, and G2 Notifications.	Brian Nguyen
13	Develop C2 Owner Web Portal atomic owner submission and idempotent publication through the C3 Backend API Service.	Kaustubh Mathur
14	Develop C5 Deal Ingestion Service durable leases, retries, duplicate matching, and G3 Configuration budget guards through C3.	Kiki Vasilev
15	Test C2 Owner Web Portal missing-field, duplicate-request, cross-venue, and submission recovery behavior.	Kaustubh Mathur
16	Develop C4 Deal Database deployment and G4 Logging and health diagnostics in the staged Azure environment.	Meredith Bartel
17	Test C1 Student Web App and G2 Notifications accessibility, keyboard navigation, and screen-reader feedback.	Brian Nguyen
18	Validate C5 Deal Ingestion Service extraction quality, permitted C6 source access, and outage and spending controls.	Kiki Vasilev
19	Test C3 Backend API Service age, geography, schedule, concurrency, and browser performance boundaries with C4 fixtures.	Ved Sanap
20	Refine C1 Student Web App and G2 Notifications defects and prepare the student demonstration and user guide.	Brian Nguyen
21	Test C4 Deal Database backup restoration, migration recovery, and G4 Logging and health failure diagnostics.	Meredith Bartel
22	Validate C2 Owner Web Portal onboarding, submit-to-feed acceptance timing, and the owner user guide.	Kaustubh Mathur
23	Validate C3 Backend API Service integrated acceptance evidence across C1-C5 and the release readiness checklist.	Ved Sanap
24	Validate C5 Deal Ingestion Service automation release approval and G3 Configuration funded settings on staging.	Kiki Vasilev
25	Document C4 Deal Database and G4 Logging and health deployment handover, final report, and demonstration evidence for C1-C6.	Meredith Bartel

Milestones List

Fall milestones are numbered F01-F13 in chronological order; spring milestones retain their S identifiers.

ID	Date / status	Milestone	Demonstration / verification	Verifier
F01	Sep 28, 2026; Complete artifact; course due date	Assignment 4 - User Stories and Use Cases	US-01 to US-04, UC-01, and AC-01.1 to AC-04.1 establish the stakeholder baseline. The artifact exists; the screenshot supplies its deadline, not a verified submission date or grade.	Brian Nguyen; instructor
F02	Oct 2, 2026; Complete artifact; course due date	Assignment 5 - High-Level System Design (Design D0)	Six components, I1-I5, primary owners, architecture and the decision log establish the design baseline. Verify against the existing D0 artifact; submission and grade are not asserted.	Ved Sanap; advisor/instructor
F03	Oct 8, 2026; Complete artifact; course due date	Assignment 6 - Detailed System Design, Part 1 (Design D1)	D1 provides the database model, algorithms, API contract and stack rationale. The document is dated October 7; the course deadline shown in the screenshot is October 8.	Meredith Bartel and Kiki Vasilev; instructor
F04	Oct 14, 2026; Complete artifact; course due date	Assignment 7 - Detailed System Design, Part 2 (Design D2)	D2 version 1.2 completes C1/C2/C6 and G1-G4, revises D1 behavior and traces requirements. The document exists; implementation/test completion and submission are not claimed.	Ved Sanap; advisor/instructor
F05	Oct 14, 2026; Complete review artifact; actions open; course due date	Assignment 7 - Peer Review Notes	Team Agri-AI feedback and PR-01 to PR-05 are recorded. Their original October 21 action dates are past; no closure evidence is supplied. Peer review shares Assignment 7's October 14 deadline.	All five members; instructor
F06	Oct 28, 2026; Prepared; course due	Assignment 8 - Task List, Milestones,	The planning package includes the separate task table, milestones, timeline, effort matrix, WBS and risk	All five members; advisor/instructor

ID	Date / status	Milestone	Demonstration / verification	Verifier
	date confirmed	Timeline, and Effort Matrix	register. Review the proposed estimates and risk ratings before submission.	
F07	Oct 30, 2026; Scheduled; course due date confirmed	Assignment 9 - Requirements Document and Technical Specification Document	The requirements and technical specification documents trace the stakeholder criteria to D2. Verify completeness and resolve conflicting requirement language using the existing design baseline.	Brian Nguyen; advisor/instructor
F08	Nov 4, 2026; Scheduled; course due date confirmed	Assignment 10 - Test and Validation Plan Draft	The draft identifies fixtures, scenarios, pass/fail rules, timing methods and price-policy verification. Check coverage of all acceptance criteria before implementation.	Kaustubh Mathur; advisor/instructor
F09	Nov 11, 2026; Scheduled; course due date confirmed	Assignment 11 - Ethics and Responsible-Use Statement	The statement addresses age self-attestation, unsafe AI classification, source permissions, privacy and venue inclusion. Review it with the security and edge-case coverage evidence.	Ved Sanap; instructor
F10	Nov 18, 2026; Scheduled; course due date confirmed	Assignment 12 - Presentation Draft and Documented Advisor Feedback	A readable presentation draft explains the design and plan; documented advisor feedback records requested revisions and the team's response. Verify the draft and feedback record together.	Meredith Bartel; advisor/instructor
F11	Nov 18, 2026; Scheduled; course due date confirmed	Assignment 13 - Team Oral Presentation, Recorded Video, and Peer Reviews	The team delivers its oral presentation, provides the recorded video and completes peer reviews. Verify the presentation materials, playable recording and review records by the shared November 18 deadline.	Kiki Vasilev and all members; instructor
F12	Dec 2, 2026; Scheduled;	Assignment 14 - Design Report, Design	The design report incorporates the current plan and presentation feedback; the	All members; advisor/instructor

ID	Date / status	Milestone	Demonstration / verification	Verifier
	course due date confirmed	Notebook, and Team and Self Assessment	notebook records design decisions and evidence; team and self assessments document contributions. Verify all required files and individual assessments.	
F13	Dec 8, 2026; Scheduled; course due date confirmed	Assignment 15 - Project Advisor Assessments	The advisor has the completed design package and contribution evidence needed for assessment. Verify that the advisor assessment is received through the course process; team preparation does not substitute for the advisor's assessment.	Dr. Hrishikesh Vinayak Bhide; instructor
S00	Jan 11, 2027; Proposed internal checkpoint	Implementation kickoff and availability baseline	Owners confirm weekly availability, access, budget, scope and calendar; actual work cannot enable automation without D2 approval evidence.	All members; advisor
S01	Jan 25, 2027; Proposed internal gate	Foundation contracts and data ready	Clean migrations, shared DTO harness, fixture UI/form and normalized adapters demonstrate that all five components can build against one contract.	Ved Sanap and Meredith Bartel; advisor
S02	Feb 9, 2027; Proposed internal gate	Secure staged services and durable worker ready	Guest/owner authorization, worker crash/retry recovery, budget pause and staging health work; live API access is demonstrated or the adapter is disabled.	Ved Sanap and Kiki Vasilev; advisor
S03	Feb 24, 2027; Proposed internal gate	Manual vertical slice and extraction evidence ready	Approved owner submits a deal visible to an eligible student; age/freshness behavior and labeled extraction evidence are demonstrated. No unsupported automation claim.	Brian Nguyen and Kaustubh Mathur; advisor
S04	Mar 10, 2027;	Boundary and accessibility	Owner failure tests, age/time/radius/concurrency	Ved Sanap and Brian Nguyen;

ID	Date / status	Milestone	Demonstration / verification	Verifier
	Proposed internal gate	evidence ready	tests, browser timing and axe/manual accessibility checks provide recorded results or explicitly block the gate.	advisor
S05	Mar 24, 2027; Proposed internal checkpoint	Post-break correction and risk checkpoint	During 20-22, review defect burn-down, owner acceptance availability and restore evidence; replan immediately if the April 5 gate is threatened.	All members; advisor
S06	Apr 5, 2027; Proposed internal gate	Acceptance candidate and recovery ready	Retested student/owner paths, onboarding guide and successful restore exercise establish a stable system for complete acceptance testing.	Meredith Bartel and Kaustubh Mathur; advisor
S07	Apr 12, 2027; Proposed internal gate	Integrated acceptance accepted	Requirements-to-results review shows passing evidence and no release-blocking age/data-integrity defects; exceptions require a recorded stakeholder scope decision.	Ved Sanap; all members and advisor
S08	Apr 19, 2027; Proposed internal gate	Automation decision and release candidate frozen	Enabled sources, funded budget, extraction evaluation and approvers are recorded; otherwise retain manual publication and explicitly report unmet automation coverage.	Kiki Vasilev; team and advisor
S09	Apr 28, 2027; Proposed internal gate	Final report and deployment handover accepted	Second-operator deployment/restore, final report, current timeline/effort matrix and known limits are reproducible from the repository.	Meredith Bartel; advisor/instructor

ID	Date / status	Milestone	Demonstration / verification	Verifier
S10	Apr 30, 2027; Proposed internal finish target	Final implementation demonstration	End-to-end student and owner walkthrough, recoverable outage, source/age restrictions and approved automation scope are demonstrated from the frozen candidate.	All members; advisor/instructor

Table 1 - Timeline

Table 1 shows each task's owner, dates, dependencies, and risks. Tasks start after their dependencies are complete, except where parallel work is noted.

Each task includes a 20% effort reserve, with additional protected time before major milestones. We will review progress weekly and update the plan when dates change.

Main spring critical path: 06 -> 11 -> 12 -> 19 -> 20 -> 23 -> 24 -> 25.

ID	Task statement	Owner	Start	Core complete	Task contingency	Finish	Dependen
01	Specify the C1 Student Web App and G2 Notifications requirements and technical specifications with acceptance traceability.	Brian Nguyen	2026-10-28	2026-10-29 14:00	Oct 29, 2026, 14:00-18:00 (4 person-hours; 20%)	2026-10-29 18:00	D2 baselin
02	Specify the C2 Owner Web Portal price policy and test/validation plan for seven-item submissions and publication timing.	Kaustubh Mathur	2026-10-29	2026-11-02	Nov 3, 2026	2026-11-03	D2 baselin parallel wit

ID	Task statement	Owner	Start	Core complete	Task contingency	Finish	Dependen
03	Refine the C3 Backend API Service and G1 Authentication and authorization contracts, edge-case coverage, and ethics analysis.	Ved Sanap	2026-11-05	2026-11-09	Nov 10, 2026	2026-11-10	01, 02
04	Document the C4 Deal Database and G4 Logging and health recovery plans, technical references, and technical communication package.	Meredith Bartel	2026-11-11	2026-11-15	Nov 16, 2026	2026-11-16	02, 03
05	Investigate C5 Deal Ingestion Service, C6 External Social Media APIs, and G3 Configuration access and release gates for the fall presentations.	Kiki Vasilev	2026-11-11	2026-11-15	Nov 16, 2026	2026-11-16	03; parallel with 04
06	Develop the C4 Deal Database migrations, relational invariants, and representative venue and deal fixtures.	Meredith Bartel	2027-01-11	2027-01-20	Jan 21, 2027 to Jan 22, 2027	2027-01-22	F13

ID	Task statement	Owner	Start	Core complete	Task contingency	Finish	Dependen
07	Develop the C3 Backend API Service shared data contracts, validation errors, and I1-I4 contract test harness.	Ved Sanap	2027-01-11	2027-01-20	Jan 21, 2027 to Jan 22, 2027	2027-01-22	F13
08	Develop the C1 Student Web App Flask/Jinja feed and detail templates using fixture-backed data.	Brian Nguyen	2027-01-11	2027-01-20	Jan 21, 2027 to Jan 22, 2027	2027-01-22	F13
09	Develop the C2 Owner Web Portal login screens and seven-item submission form with accessible field validation.	Kaustubh Mathur	2027-01-11	2027-01-20	Jan 21, 2027 to Jan 22, 2027	2027-01-22	F13
10	Develop the C5 Deal Ingestion Service normalized C6 External Social Media APIs adapters and replayable post fixtures.	Kiki Vasilev	2027-01-11	2027-01-20	Jan 21, 2027 to Jan 22, 2027	2027-01-22	F13
11	Develop C3 Backend API Service and G1 Authentication	Ved Sanap	2027-01-26	2027-02-04	Feb 5, 2027 to Feb 6, 2027	2027-02-06	06, 07, S0

ID	Task statement	Owner	Start	Core complete	Task contingency	Finish	Dependen
	and authorization guest age sessions and verified owner access.						
12	Develop C1 Student Web App live feed integration, pagination, source and freshness displays, and G2 Notifications.	Brian Nguyen	2027-02-10	2027-02-19	Feb 20, 2027 to Feb 21, 2027	2027-02-21	08, 11, S02
13	Develop C2 Owner Web Portal atomic owner submission and idempotent publication through the C3 Backend API Service.	Kaustubh Mathur	2027-02-10	2027-02-19	Feb 20, 2027 to Feb 21, 2027	2027-02-21	09, 11, S02
14	Develop C5 Deal Ingestion Service durable leases, retries, duplicate matching, and G3 Configuration budget guards through C3.	Kiki Vasilev	2027-01-26	2027-02-04	Feb 5, 2027 to Feb 6, 2027	2027-02-06	06, 07, 10, S01
15	Test C2 Owner Web Portal missing-field, duplicate-request, cross-	Kaustubh Mathur	2027-02-25	2027-03-06	Mar 7, 2027 to Mar 8, 2027	2027-03-08	13, S03

ID	Task statement	Owner	Start	Core complete	Task contingency	Finish	Dependen
	venue, and submission recovery behavior.						
16	Develop C4 Deal Database deployment and G4 Logging and health diagnostics in the staged Azure environment.	Meredith Bartel	2027-01-26	2027-02-04	Feb 5, 2027 to Feb 6, 2027	2027-02-06	06, 07, S01
17	Test C1 Student Web App and G2 Notifications accessibility, keyboard navigation, and screen-reader feedback.	Brian Nguyen	2027-02-25	2027-03-06	Mar 7, 2027 to Mar 8, 2027	2027-03-08	12, 16, S03
18	Validate C5 Deal Ingestion Service extraction quality, permitted C6 source access, and outage and spending controls.	Kiki Vasilev	2027-02-10	2027-02-19	Feb 20, 2027 to Feb 21, 2027	2027-02-21	05, 14, 16, S02
19	Test C3 Backend API Service age, geography, schedule, concurrency, and browser performance boundaries	Ved Sanap	2027-02-25	2027-03-06	Mar 7, 2027 to Mar 8, 2027	2027-03-08	11, 12, 13, 16, S03

ID	Task statement	Owner	Start	Core complete	Task contingency	Finish	Dependen
	with C4 fixtures.						
20	Refine C1 Student Web App and G2 Notifications defects and prepare the student demonstration and user guide.	Brian Nguyen	2027-03-22	2027-03-31	Apr 1, 2027 to Apr 2, 2027	2027-04-02	17, 19, S04
21	Test C4 Deal Database backup restoration, migration recovery, and G4 Logging and health failure diagnostics.	Meredith Bartel	2027-03-22	2027-03-31	Apr 1, 2027 to Apr 2, 2027	2027-04-02	16, 19, S04
22	Validate C2 Owner Web Portal onboarding, submit-to-feed acceptance timing, and the owner user guide.	Kaustubh Mathur	2027-03-22	2027-03-31	Apr 1, 2027 to Apr 2, 2027	2027-04-02	15, 17, 19, S04
23	Validate C3 Backend API Service integrated acceptance evidence across C1-C5 and the release readiness checklist.	Ved Sanap	2027-04-06	2027-04-10	Apr 11, 2027	2027-04-11	18, 20, 21, S06

ID	Task statement	Owner	Start	Core complete	Task contingency	Finish	Dependen
24	Validate C5 Deal Ingestion Service automation release approval and G3 Configuration funded settings on staging.	Kiki Vasilev	2027-04-13	2027-04-17	Apr 18, 2027	2027-04-18	18, 21, 23, S07
25	Document C4 Deal Database and G4 Logging and health deployment handover, final report, and demonstration evidence for C1-C6.	Meredith Bartel	2027-04-20	2027-04-24	Apr 25, 2027	2027-04-25	23, 24, S08

Table 1 continued - milestone dates and protected blocks

ID	Milestone	Milestone date / course deadline	Incoming dependencies	Protected block
F01	Assignment 4 - User Stories and Use Cases	2026-09-28	Existing artifact	Historical deadline
F02	Assignment 5 - High-Level System Design (Design D0)	2026-10-02	Existing artifact	Historical deadline
F03	Assignment 6 - Detailed System Design, Part 1 (Design D1)	2026-10-08	Existing artifact	Historical deadline
F04	Assignment 7 - Detailed System Design, Part 2 (Design D2)	2026-10-14	Existing artifact	Historical deadline

ID	Milestone	Milestone date / course deadline	Incoming dependencies	Protected block
F05	Assignment 7 - Peer Review Notes	2026-10-14	Existing artifact	Historical deadline
F06	Assignment 8 - Task List, Milestones, Timeline, and Effort Matrix	2026-10-28	Calendar checkpoint	Due today; no retrospective buffer asserted
F07	Assignment 9 - Requirements Document and Technical Specification Document	2026-10-30	01	2026-10-30
F08	Assignment 10 - Test and Validation Plan Draft	2026-11-04	02	2026-11-04
F09	Assignment 11 - Ethics and Responsible-Use Statement	2026-11-11	03	2026-11-11
F10	Assignment 12 - Presentation Draft and Documented Advisor Feedback	2026-11-18	04	2026-11-17 to 2026-11-18
F11	Assignment 13 - Team Oral Presentation, Recorded Video, and Peer Reviews	2026-11-18	04, 05	2026-11-17 to 2026-11-18 (shared with F10)
F12	Assignment 14 - Design Report, Design Notebook, and Team and Self Assessment	2026-12-02	F10, F11; 01-05 evidence	2026-11-30 to 2026-12-02
F13	Assignment 15 - Project Advisor Assessments	2026-12-08	F12; advisor assessment	2026-12-03 to 2026-12-07
S00	Implementation kickoff and availability baseline	2027-01-11	F13	Not a future major gate
S01	Foundation contracts and data ready	2027-01-25	06, 07, 08, 09, 10	2027-01-23 to 2027-01-25
S02	Secure staged services and durable worker ready	2027-02-09	11, 14, 16	2027-02-07 to 2027-02-09
S03	Manual vertical slice and extraction evidence ready	2027-02-24	12, 13, 18	2027-02-22 to 2027-02-24

ID	Milestone	Milestone date / course deadline	Incoming dependencies	Protected block
S04	Boundary and accessibility evidence ready	2027-03-10	15, 17, 19	2027-03-09 to 2027-03-10
S05	Post-break correction and risk checkpoint	2027-03-24	Calendar checkpoint	Not a future major gate
S06	Acceptance candidate and recovery ready	2027-04-05	20, 21, 22	2027-04-03 to 2027-04-05
S07	Integrated acceptance accepted	2027-04-12	23	2027-04-12
S08	Automation decision and release candidate frozen	2027-04-19	24	2027-04-19
S09	Final report and deployment handover accepted	2027-04-28	25	2027-04-26 to 2027-04-28
S10	Final implementation demonstration	2027-04-30	S09	2027-04-29 to 2027-04-30

Availability blocks

ID	Start	End	Constraint
B01	2026-11-26	2026-11-30	Protected holiday / reduced-availability reserve; no new planned task work.
B02	2026-12-09	2027-01-10	Protected finals / winter pause; no implementation work is assumed.
B03	2027-03-11	2027-03-21	Protected exam / spring reduced-availability reserve; no new task work.

F10 and F11 are both due November 18 and share a protected November 17-18 review/submission block. F12 is due December 2, and F13 is due December 8. Break blocks remain planning assumptions rather than official university dates.

Table 2 - Effort Matrix

Table 2 shows each member's estimated hours, including 20% contingency. An asterisk (*) marks the primary owner, who carries 60% of the task effort; two contributors carry 20% each.

Total planned effort is **894 hours**: 745 base hours plus 149 contingency hours. These estimates will be reviewed by the team.

ID	Task statement	Brian Nguyen	Kaustubh Mathur	Ved Sanap	Meredith Bartel	Kiki Vasilev	Total
01	Specify the C1 Student Web App and G2 Notifications requirements and technical specifications with acceptance traceability.	14.4*	4.8	0	0	4.8	24
02	Specify the C2 Owner Web Portal price policy and test/validation plan for seven-item submissions and publication timing.	4.8	14.4*	0	0	4.8	24
03	Refine the C3 Backend API Service and G1 Authentication and authorization contracts, edge-case coverage, and ethics analysis.	0	0	14.4*	4.8	4.8	24
04	Document the C4 Deal Database and G4 Logging and health recovery plans, technical references, and technical communication package.	4.8	4.8	0	14.4*	0	24
05	Investigate C5 Deal Ingestion Service, C6 External Social	0	0	4.8	4.8	14.4*	24

ID	Task statement	Brian Nguyen	Kaustubh Mathur	Ved Sanap	Meredith Bartel	Kiki Vasilev	Total
	Media APIs, and G3 Configuration access and release gates for the fall presentations.						
06	Develop the C4 Deal Database migrations, relational invariants, and representative venue and deal fixtures.	7.2	0	7.2	21.6*	0	36
07	Develop the C3 Backend API Service shared data contracts, validation errors, and I1-I4 contract test harness.	7.2	0	21.6*	0	7.2	36
08	Develop the C1 Student Web App Flask/Jinja feed and detail templates using fixture-backed data.	21.6*	7.2	0	7.2	0	36
09	Develop the C2 Owner Web Portal login screens and seven-item submission form with accessible field validation.	7.2	21.6*	0	0	7.2	36
10	Develop the C5 Deal Ingestion Service normalized C6 External Social Media APIs adapters and	0	0	7.2	7.2	21.6*	36

ID	Task statement	Brian Nguyen	Kaustubh Mathur	Ved Sanap	Meredith Bartel	Kiki Vasilev	Total
	replayable post fixtures.						
11	Develop C3 Backend API Service and G1 Authentication and authorization guest age sessions and verified owner access.	9.6	9.6	28.8*	0	0	48
12	Develop C1 Student Web App live feed integration, pagination, source and freshness displays, and G2 Notifications.	28.8*	0	9.6	9.6	0	48
13	Develop C2 Owner Web Portal atomic owner submission and idempotent publication through the C3 Backend API Service.	0	28.8*	9.6	9.6	0	48
14	Develop C5 Deal Ingestion Service durable leases, retries, duplicate matching, and G3 Configuration budget guards through C3.	0	0	9.6	9.6	28.8*	48
15	Test C2 Owner Web Portal missing-field, duplicate-request, cross-venue, and	6	18*	0	0	6	30

ID	Task statement	Brian Nguyen	Kaustubh Mathur	Ved Sanap	Meredith Bartel	Kiki Vasilev	Total
	submission recovery behavior.						
16	Develop C4 Deal Database deployment and G4 Logging and health diagnostics in the staged Azure environment.	0	0	8.4	25.2*	8.4	42
17	Test C1 Student Web App and G2 Notifications accessibility, keyboard navigation, and screen-reader feedback.	21.6*	7.2	0	7.2	0	36
18	Validate C5 Deal Ingestion Service extraction quality, permitted C6 source access, and outage and spending controls.	8.4	8.4	0	0	25.2*	42
19	Test C3 Backend API Service age, geography, schedule, concurrency, and browser performance boundaries with C4 fixtures.	0	0	28.8*	9.6	9.6	48
20	Refine C1 Student Web App and G2 Notifications defects and prepare the student demonstration and user guide.	18*	6	0	6	0	30

ID	Task statement	Brian Nguyen	Kaustubh Mathur	Ved Sanap	Meredith Bartel	Kiki Vasilev	Total
21	Test C4 Deal Database backup restoration, migration recovery, and G4 Logging and health failure diagnostics.	7.2	0	0	21.6*	7.2	36
22	Validate C2 Owner Web Portal onboarding, submit-to-feed acceptance timing, and the owner user guide.	6	18*	0	6	0	30
23	Validate C3 Backend API Service integrated acceptance evidence across C1-C5 and the release readiness checklist.	8.4	0	25.2*	0	8.4	42
24	Validate C5 Deal Ingestion Service automation release approval and G3 Configuration funded settings on staging.	0	6	6	0	18*	30
25	Document C4 Deal Database and G4 Logging and health deployment handover, final report, and demonstration evidence for C1-C6.	7.2	7.2	0	21.6*	0	36
TOTAL		188.4	162	181.2	186	176.4	894

Each member has 162-188.4 planned hours, within 10% of the 178.8-hour average. Weekly planning limits are 18 hours per member in fall and 30 during implementation, subject to confirmed availability. We will review workloads at each milestone and transfer supporting work if someone becomes overloaded, while keeping primary responsibilities clear. The timeline and effort matrix will be updated together.

Three-point estimate inputs (proposed person-hours)

ID	O	M	P	Weighted E	20% reserve	Budget
01	15	20	25	20	4	24
02	15	20	25	20	4	24
03	15	20	25	20	4	24
04	15	20	25	20	4	24
05	15	20	25	20	4	24
06	22.5	30	37.5	30	6	36
07	22.5	30	37.5	30	6	36
08	22.5	30	37.5	30	6	36
09	22.5	30	37.5	30	6	36
10	22.5	30	37.5	30	6	36
11	30	40	50	40	8	48
12	30	40	50	40	8	48
13	30	40	50	40	8	48
14	30	40	50	40	8	48
15	18.8	25	31.2	25	5	30
16	26.2	35	43.8	35	7	42
17	22.5	30	37.5	30	6	36
18	26.2	35	43.8	35	7	42
19	30	40	50	40	8	48
20	18.8	25	31.2	25	5	30
21	22.5	30	37.5	30	6	36
22	18.8	25	31.2	25	5	30
23	26.2	35	43.8	35	7	42
24	18.8	25	31.2	25	5	30
25	22.5	30	37.5	30	6	36

Work breakdown structure

BarCat Deals is the root. Numbered phases are level 1; task IDs are level 2. Each task appears once. Historical artifacts are baseline inputs, not additional implementation tasks.

1. Planning and course deliverables

1. Specify the C1 Student Web App and G2 Notifications requirements and technical specifications with acceptance traceability. Owner: Brian Nguyen.
2. Specify the C2 Owner Web Portal price policy and test/validation plan for seven-item submissions and publication timing. Owner: Kaustubh Mathur.
3. Refine the C3 Backend API Service and G1 Authentication and authorization contracts, edge-case coverage, and ethics analysis. Owner: Ved Sanap.
4. Document the C4 Deal Database and G4 Logging and health recovery plans, technical references, and technical communication package. Owner: Meredith Bartel.
5. Investigate C5 Deal Ingestion Service, C6 External Social Media APIs, and G3 Configuration access and release gates for the fall presentations. Owner: Kiki Vasilev.

2. Implementation foundations

6. Develop the C4 Deal Database migrations, relational invariants, and representative venue and deal fixtures. Owner: Meredith Bartel.
7. Develop the C3 Backend API Service shared data contracts, validation errors, and I1-I4 contract test harness. Owner: Ved Sanap.
8. Develop the C1 Student Web App Flask/Jinja feed and detail templates using fixture-backed data. Owner: Brian Nguyen.
9. Develop the C2 Owner Web Portal login screens and seven-item submission form with accessible field validation. Owner: Kaustubh Mathur.
10. Develop the C5 Deal Ingestion Service normalized C6 External Social Media APIs adapters and replayable post fixtures. Owner: Kiki Vasilev.

3. Integrated feature development

11. Develop C3 Backend API Service and G1 Authentication and authorization guest age sessions and verified owner access. Owner: Ved Sanap.
12. Develop C1 Student Web App live feed integration, pagination, source and freshness displays, and G2 Notifications. Owner: Brian Nguyen.
13. Develop C2 Owner Web Portal atomic owner submission and idempotent publication through the C3 Backend API Service. Owner: Kaustubh Mathur.
14. Develop C5 Deal Ingestion Service durable leases, retries, duplicate matching, and G3 Configuration budget guards through C3. Owner: Kiki Vasilev.

15. Develop C4 Deal Database deployment and G4 Logging and health diagnostics in the staged Azure environment. Owner: Meredith Bartel.

4. Verification and defect correction

15. Test C2 Owner Web Portal missing-field, duplicate-request, cross-venue, and submission recovery behavior. Owner: Kaustubh Mathur.
16. Test C1 Student Web App and G2 Notifications accessibility, keyboard navigation, and screen-reader feedback. Owner: Brian Nguyen.
17. Validate C5 Deal Ingestion Service extraction quality, permitted C6 source access, and outage and spending controls. Owner: Kiki Vasilev.
18. Test C3 Backend API Service age, geography, schedule, concurrency, and browser performance boundaries with C4 fixtures. Owner: Ved Sanap.
19. Refine C1 Student Web App and G2 Notifications defects and prepare the student demonstration and user guide. Owner: Brian Nguyen.
20. Test C4 Deal Database backup restoration, migration recovery, and G4 Logging and health failure diagnostics. Owner: Meredith Bartel.
21. Validate C2 Owner Web Portal onboarding, submit-to-feed acceptance timing, and the owner user guide. Owner: Kaustubh Mathur.

5. Acceptance and release

23. Validate C3 Backend API Service integrated acceptance evidence across C1-C5 and the release readiness checklist. Owner: Ved Sanap.
24. Validate C5 Deal Ingestion Service automation release approval and G3 Configuration funded settings on staging. Owner: Kiki Vasilev.
25. Document C4 Deal Database and G4 Logging and health deployment handover, final report, and demonstration evidence for C1-C6. Owner: Meredith Bartel.

Risk register

Ratings and strategies are proposed interpretations of the D2/lecture risks, not a record of team approval. Likelihood: Low = unlikely; Medium = plausible; High = unresolved prerequisite or observed concern. Impact: High blocks a release requirement/gate, Medium needs local rework, Low is cosmetic. Current open actions are issues; these rows describe their possible downstream consequences.

R1 - Permitted social-post access is unavailable or revoked

Category: External. Likelihood: High. Impact: High. Strategy: Reduce. Owner: Kiki Vasilev.

Basis: D2 OQ-01, DR-08, EC-14 and open PR-02 already identify unconfirmed access.

Mitigation: Verify one permitted fetch per enabled adapter, honor throttling, disable unapproved accounts, retain fixtures for development, and keep the owner-only venue path usable; do not claim full automated coverage.

Trigger: No permitted-fetch evidence at S01, or any 401/403 revocation / sustained 429 after access is enabled.

Timeline linkage: 05, 10, 14, 18, 24; S01/S02/S08 protected blocks

R2 - AI extraction invents or misclassifies an offer

Category: Technical. Likelihood: High. Impact: High. Strategy: Reduce. Owner: Kiki Vasilev.

Basis: D2 OQ-04, EC-12, DR-07/DR-09 and PR-05 leave quality approval open.

Mitigation: Keep AI publication off until the team-approved labeled evaluation passes; hold uncertain time/price/source records, treat uncertain alcohol conservatively, validate complete batches and preserve manual submission.

Trigger: Any invented fact or unsafe alcohol classification in the agreed evaluation, or no accepted checklist by F12.

Timeline linkage: 05, 14, 18, 23, 24; S03/S07/S08 blocks

R3 - Age or owner authorization fails, or access is unusable

Category: Technical. Likelihood: Medium. Impact: High. Strategy: Reduce. Owner: Ved Sanap.

Basis: D2 EC-01/EC-04/EC-15 and AC-04.1 identify sensitive authorization and accessibility boundaries.

Mitigation: Apply one server eligibility rule to feed/detail, verified owner mapping and CSRF; exercise expired/revoked sessions, cross-venue writes, direct requests and assistive-technology paths. Block release for an age leak.

Trigger: Any unauthorized alcohol/venue content or serious/critical accessibility result at S04 or acceptance.

Timeline linkage: 03, 11-13, 15, 17, 19, 20, 22, 23; S04/S07 blocks

R4 - D1/D2 drift, schedule boundaries or duplicate writes break integration

Category: Technical. Likelihood: Medium. Impact: High. Strategy: Reduce. Owner: Ved Sanap.

Basis: D2 explicitly replaces active-only default behavior and documents midnight, radius, price-policy and idempotency gaps.

Mitigation: Use D2 contracts, approved coordinates and price policy; migrate fixtures first; test midnight/DST, one-mile and 24-hour boundaries, pagination, lost responses and simultaneous workers. Freeze interfaces at S01.

Trigger: Contract mismatch at S01/S03, unresolved OQ-02/OQ-03, or a duplicate/partial publication in 19.

Timeline linkage: 01, 02, 06, 07, 12-15, 19, 22, 23; S01/S03/S04 blocks

R5 - Database, migration or deployment failure loses service or evidence

Category: Technical. Likelihood: Medium. Impact: High. Strategy: Reduce. Owner: Meredith Bartel.

Basis: D2 Section 2.4 and EC-08 to EC-11 require atomic transactions, backups and recoverable work.

Mitigation: Validate migration/readiness, unique constraints, venue locks and lease recovery; confirm hosting backup settings, run a restore exercise and retain a second-operator runbook before acceptance.

Trigger: Failed migration/restore, missing backup capability or stale worker completion accepted during tests.

Timeline linkage: 04, 06, 14, 16, 19, 21, 25; S02/S06/S09 blocks

R6 - Hosting, identity or AI costs and quotas exceed available funding

Category: External. Likelihood: Medium. Impact: High. Strategy: Reduce. Owner: Kiki Vasilev.

Basis: D2 OQ-04 and the economic constraint leave funded caps and hosting capabilities unverified.

Mitigation: Confirm funded limits and terms, reserve capped AI cost through C3, monitor usage, test cap pause, and disable optional automation when unfunded. Do not assume student credits or free-tier capacity.

Trigger: No funded cap at S02, projected spend reaches the team-approved alert level, or quota/service readiness fails.

Timeline linkage: 05, 14, 16, 18, 21, 24; S02/S08 blocks